A FOURTH HAT: BIPARTITE GRAPHS OF MAXIMUM DEGREE $q+2$ AND DIAMETER 3 ON $2(q^2+q+6)$ VERTICES

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Abstract. For every projective plane of order $q \geq 4$ we construct a bipartite graph with $2(q^2+q+6)$ vertices, maximum degree $q+2$ and diameter 3. This is two vertices more than the family of Araujo-Pardo and López, and for $q = 9, 11, 13$ it gives bipartite $(11,3)$-, $(13,3)$- and $(15,3)$-graphs of orders 192, 276 and 376, each two more than the largest previously listed. The construction adds a fourth "hat" pair to a projective plane and makes room for it by rotating three incidences. The proof uses only the axioms of a projective plane and a self-duality of the construction.

1. Introduction

A *bipartite* (Δ, D)*-graph* is a bipartite graph of maximum degree at most Δ and diameter at most D. The bipartite degree/diameter problem asks for the largest order of such a graph; see the survey of Miller and Širáň [4] and the table maintained on the Combinatorics Wiki [5]. For $D = 3$ the bipartite Moore bound is $2(\Delta^2 - \Delta + 1)$, and it is attained by the point–line incidence graph of a projective plane of order $\Delta - 1$. When $\Delta - 1$ is not the order of a known projective plane, the best known graphs are built by modifying a plane of a smaller order.

For $\Delta = q + 2$ with q a prime power, Araujo-Pardo and López [2] add a small number of vertices to the incidence graph of $\mathrm{PG}(2, q)$ and obtain bipartite $(q+2, 3)$-graphs of order $2(q^2+q+5)$. For $\Delta = 11$ the same order 190 was found independently by Feria-Purón, Miller and Pineda-Villavicencio [3]. These are the listed values for $(11,3)$, $(13,3)$ and $(15,3)$ in [5], and the paper [2] asks for improvements of the construction.

Theorem 1. *Let* Π *be a projective plane of order* $q \geq 4$*. There is a bipartite graph of order* $2(q^2+q+6)$*, maximum degree* $q+2$ *and diameter* 3*, whose two sides extend the points and the lines of* Π*.*

The plane need not be Desarguesian. Table 1 lists the orders where the theorem matters, namely those $\Delta = q + 2$ with q a prime power and $\Delta - 1$ not a prime power (when $\Delta - 1$ is a prime power, $\mathrm{PG}(2, \Delta - 1)$ already meets the Moore bound).

The graphs for $q = 9, 11, 13$ were first found by a computer search (Appendix A); the general rule below was then read off from them.

2. The construction

Let Π be a projective plane of order $q \geq 4$. Fix

- a line ℓ and a point $A \notin \ell$;
- a point $P \in \ell$;
- four distinct points $X_0, X_1, X_a, X_c \in \ell \setminus \{P\}$ (possible since $|\ell| = q + 1 \geq 5$).

Date: October 2026. Preliminary note; not yet refereed.

Δ	q	$2(q^2+q+5)$	$2(q^2+q+6)$	Moore bound	largest listed in [5]
7	5	70	72	86	80
11	9	190	**192**	222	190
13	11	274	**276**	314	274
15	13	374	**376**	422	374
19	17	622	624	686	—
21	19	770	772	842	—
25	23	1114	1116	1202	—

TABLE 1. Orders given by Theorem 1 for $\Delta \le 25$. The table in [5] stops at $\Delta = 16$; for $\Delta = 7$ a larger graph is known.

For $X \in \ell$ write AX for the line through A and X; these are the $q+1$ lines through A, and $AX \cap \ell = \{X\}$. Let

$$\ell^* = \ell \setminus \{X_0, X_1, X_c\},$$

a set of $q-2$ points containing P and X_a. Write $S = \{0, 1, a, c\}$.

Add five new points Q, H_0, H_1, H_a, H_c and five new lines M, K_0, K_1, K_a, K_c. The incidence structure $\mathcal{S}$ has the following incidences.

(R1) Every incidence of Π except (X_s, AX_s) for $s \in S$; and, in addition, the *rotation*

$$X_1 \in AX_0, \qquad X_c \in AX_1, \qquad X_0 \in AX_c.$$

(R2) For each $s \in S$, H_s lies on every line of Π through X_s except ℓ.
(R3) For each $s \in S$, K_s contains every point of AX_s except A.
(R4) M contains the points of ℓ^* and H_0, H_1, H_a, H_c.
(R5) Q lies on AX for every $X \in \ell^*$, and on K_0, K_1, K_a, K_c.
(R6) $A \in K_a$, $H_a \in \ell$, $H_0 \in K_c$, $H_c \in K_1$, $H_1 \in K_0$.

Let G be the incidence graph of $\mathcal{S}$: its vertices are the points and lines of $\mathcal{S}$, and a point is adjacent to a line when they are incident. G is bipartite by definition, and it has $2(q^2+q+1+5) = 2(q^2+q+6)$ vertices.

Informally, H_s is a copy of the point X_s without ℓ, and K_s is a copy of the line AX_s without A; a pair (H_s, K_s) is a *hat*. The construction of [2] uses three hats. Here a fourth hat is added, and the rotation in (R1) supplies the extra incidences it needs.

3. Proof of Theorem 1

Lemma 2 (Degrees)**.** *Every vertex of G has degree at most $q+2$.*

Proof. In Π every point and every line has degree $q+1$. We list, for each vertex, what (R1)–(R6) remove and add.

Points of Π.

- A point $u \notin \ell$, $u \neq A$, loses nothing. It gains K_s exactly when $u \in AX_s$; since u lies on only one line through A, it gains at most one line. Degree $\le q+2$.
- A gains only K_a. Degree $q+2$.
- $X \in \ell^* \setminus \{X_a\}$ gains only M (it lies on no AX_s, since $AX_s \cap \ell = \{X_s\}$). Degree $q+2$.
- X_a loses AX_a and gains M and K_a. Degree $q+2$.
- X_0 loses AX_0, gains AX_c and K_0. Likewise X_1 loses AX_1 and gains AX_0 and K_1, and X_c loses AX_c and gains AX_1 and K_c. Degree $q+2$ each.

New points. Q lies on $q-2$ lines AX and on four lines K_s: degree $q+2$. Each H_s lies on q lines of Π by (R2), on M, and on exactly one line from (R6) (ℓ, K_c, K_1 or K_0): degree $q+2$.

Lines of Π.

- A line $L \neq \ell$ not through A loses nothing, since every removed incidence is on a line through A. It gains H_s exactly when $X_s \in L$, and L meets ℓ in one point, so it gains at most one hat. Degree $\leq q+2$.
- ℓ gains only H_a. Degree $q+2$.
- AX with $X \in \ell^* \setminus \{X_a\}$ (this includes AP) gains only Q. Degree $q+2$.
- AX_a loses X_a and gains Q and H_a. Degree $q+2$.
- AX_0 loses X_0, gains X_1 and H_0. Likewise AX_1 loses X_1 and gains X_c and H_1, and AX_c loses X_c and gains X_0 and H_c. Degree $q+2$ each.

New lines. M contains $q-2$ points of ℓ^* and four hats: degree $q+2$. Each K_s contains q points of $AX_s \setminus \{A\}$, the point Q, and exactly one point from (R6) (A, H_c, H_0 or H_1 for $s = a, 1, c, 0$ respectively): degree $q+2$. □

The diameter bound reduces to two covering properties.

Lemma 3. *If any two points of $\mathcal{S}$ lie on a common line and any two lines of $\mathcal{S}$ meet in a common point, then G has diameter at most 3.*

Proof. Two points, or two lines, are at distance 2. Let p be a point and L a line not incident with p. Pick any point $p' \in L$ and a line L' through p and p'; then p, L', p', L is a path of length 3. □

Lemma 4 (Points)**.** *Any two points of $\mathcal{S}$ lie on a common line of $\mathcal{S}$.*

Proof. Only the four incidences (X_s, AX_s) of Π are removed.

(a) Two points u, v of Π. Let $L = uv$ in Π. If both incidences survive, L works. Otherwise $L = AX_s$ and, say, $u = X_s$. If $v \neq A$ then $u, v \in AX_s \setminus \{A\} \subseteq K_s$. If $v = A$, use K_a for $s = a$, and the rotation for the others: X_0 and A lie on AX_c, X_1 and A on AX_0, X_c and A on AX_1.

(b) Q and a point u of Π. If $u = A$, use AP (as $P \in \ell^*$). If $u \in \ell^* \setminus \{X_a\}$, use Au. If $u = X_a$, use K_a. If $u = X_s$ with $s \in \{0, 1, c\}$, use K_s. If $u \notin \ell$ and $u \neq A$, let $X = Au \cap \ell$: if $X \in \ell^*$ use AX; otherwise $X = X_s$ with $s \in \{0, 1, c\}$, and $u \in AX_s \setminus \{A\} \subseteq K_s$.

(c) H_s and a point u of Π. If $u \notin \ell$ (including $u = A$), the line uX_s of Π is not ℓ, so it contains H_s by (R2), and u keeps all its incidences. If $u = X_s$, use any line through X_s other than ℓ and AX_s. If $u \in \ell^*$, use M. Finally let $u \in \{X_0, X_1, X_c\}$, $u \neq X_s$:

- $s = a$: use ℓ.
- $s = 0$: X_1 via AX_0 (rotation); X_c via K_c.
- $s = 1$: X_c via AX_1 (rotation); X_0 via K_0.
- $s = c$: X_0 via AX_c (rotation); X_1 via K_1.

(d) Two new points. Two hats share M. Finally Q and H_0 share K_c, Q and H_c share K_1, Q and H_1 share K_0, and Q and H_a share AX_a (as $X_a \in \ell^*$). □

Lemma 5 (Self-duality)**.** *Apply the construction of Section 2 to the dual plane Π^* with the choices*

$$\ell' = A, \quad A' = \ell, \quad P' = AP, \quad X'_0 = AX_0,\ X'_1 = AX_c,\ X'_a = AX_a,\ X'_c = AX_1.$$

The resulting structure is the dual of $\mathcal{S}$, under the correspondence

$$\begin{aligned} H_0' = K_0, \quad H_1' = K_c, \quad H_a' = K_a, \quad H_c' = K_1, \\ K_0' = H_0, \quad K_1' = H_c, \quad K_a' = H_a, \quad K_c' = H_1, \\ Q' = M, \quad M' = Q. \end{aligned}$$

Proof. The choices are valid: $\ell \not\ni A$, and $AP, AX_0, AX_1, AX_a, AX_c$ are five distinct lines through A. In Π^* the line through A' and X', for $X' = AX$, is the point $AX \cap \ell = X$. Hence $A'X_0' = X_0$, $A'X_1' = X_c$, $A'X_a' = X_a$, $A'X_c' = X_1$, and ℓ'^* is the set of lines AX with $X \in \ell^*$. We translate each rule back to Π.

- (R1): the removed incidences are again (X_s, AX_s), $s \in S$. The rotation "X_1' on $A'X_0'$, X_c' on $A'X_1'$, X_0' on $A'X_c'$" reads $X_0 \in AX_c$, $X_c \in AX_1$, $X_1 \in AX_0$: the same three incidences.
- (R2): H_s' contains every point of Π on the line X_s' except A; for $X_s' = AX_t$ this is K_t.
- (R3): K_s' lies on every line of Π through the point $A'X_s' = X_t$ except ℓ; this is H_t.
- (R4): M' lies on the lines AX, $X \in \ell^*$, and on all K_t; this is Q.
- (R5): Q' contains the points $X \in \ell^*$ and all H_t; this is M.
- (R6): "$A' \in K_a'$, $H_a' \in \ell'$, $H_0' \in K_c'$, $H_c' \in K_1'$, $H_1' \in K_0'$" reads $H_a \in \ell$, $A \in K_a$, $H_1 \in K_0$, $H_0 \in K_c$, $H_c \in K_1$: the same five incidences. □

Proof of Theorem 1. G is bipartite with $2(q^2+q+6)$ vertices, and its maximum degree is at most $q+2$ by Lemma 2 (and equals $q+2$, e.g. at A). By Lemma 4, any two points of $\mathcal{S}$ are collinear. By Lemma 5, the dual of $\mathcal{S}$ is an instance of the same construction in the plane Π^* of order q, so Lemma 4 applied there shows that any two lines of $\mathcal{S}$ meet. Lemma 3 gives diameter at most 3, and the diameter is exactly 3 because G is bipartite and not complete bipartite. □

Remark 6. The graph G need not be regular. In every case we computed, adding a maximum matching between the points and lines of degree less than $q+2$ that are not yet adjacent makes it $(q+2)$-regular; adding edges cannot increase the diameter.

Remark 7 (Limits of the method). We looked for +3 and +4 inside the same family: $\Pi = \mathrm{PG}(2,q)$ is kept fixed outside a small gadget around ℓ and A, more hats, copies of A or copies of ℓ are added, and every incidence inside the gadget is left free. Every such SAT instance we tried, for q between 5 and 13, is unsatisfiable. This rules out only extensions of this specific shape, not larger graphs in general.

Appendix A. How the construction was found and checked

The graphs were found by an evolutionary program search, in the spirit of AlphaEvolve [1] and following Xu's use of Claude Haiku as the mutator for square packing [6], in which a language model (Claude Haiku, by Anthropic) writes candidate Python programs that build graphs, and an independent checker scores them. The search produced bipartite $(11,3)$-, $(13,3)$- and $(15,3)$-graphs of orders 192, 276 and 376. Comparing them with the Araujo-Pardo–López graphs showed that, outside a small gadget around ℓ and A and apart from filler edges, each is a projective plane with four hats; the three gadgets differed, and the clean rotation of Section 2 was extracted from the $q = 9$ graph, which uses exactly that rotation. The proof above was drafted with the help of Claude (Anthropic) and then checked as follows.

- *Lean.* Theorem 1 is formalized in Lean 4 with Mathlib (`plus2` and `plus2_of_order`), for every finite projective plane; `#print axioms` lists only the three standard axioms, and an independent rebuild from source gave the same result.

- *Three implementations.* The construction was implemented three times from its description, once by the agent that extracted the rule and twice independently from the six rules alone. All three agree.
- *All inputs for small q.* For $q = 4, 5, 7, 8, 9$, every ordered choice of (X_0, X_1, X_a, X_c) in $\mathrm{PG}(2, q)$ gives a valid graph (24, 120, 840, 1680 and 3024 choices). Since $\mathrm{PGL}(3, q)$ is transitive on the choices of (A, ℓ, P), this covers every input in these planes. For $q = 11, 13$, 300 random choices each; for $q = 16$, three random frames.
- *Non-Desarguesian planes.* 200 random frames in each of two non-Desarguesian planes of order 9, coordinatised by the Dickson nearfield of order 9 with its product taken in either order.
- *Three checkers* (a bitset diameter check, a check built on `networkx`, and a from-scratch breadth-first search from every vertex) accept every graph above and reject deliberately damaged copies.
- *Negative controls.* Dropping the rotation, or dropping the incidence $H_1 \in K_0$, gives diameter greater than 3 in every case tested.

The literature check behind the "largest listed" column of Table 1 searched arXiv, Crossref, OpenAlex, Google Scholar and Semantic Scholar, and the edit history of [5], in October 2026. Code, graph files and the Lean proofs are at `https://github.com/dogaozden/degree-diameter`.

References

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